

7.1 2020/0809 이종민

$$8. 3A + 2B = \begin{bmatrix} 3 & -6 & 15 \\ 12 & 12 & 24 \\ -9 & 3 & 0 \end{bmatrix} + \begin{bmatrix} 10 & 4 & 0 \\ -10 & 6 & -8 \\ -8 & 4 & -8 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & -2 & 15 \\ 22 & 18 & 16 \\ -17 & 7 & -8 \end{bmatrix}$$

$$2B + 3A = \begin{bmatrix} 13 & -2 & 15 \\ 22 & 18 & 16 \\ -17 & 7 & -8 \end{bmatrix} = 3A + 2B$$

$$0A + B = B = \begin{bmatrix} 5 & 2 & 0 \\ -5 & 3 & -4 \\ -4 & 2 & -4 \end{bmatrix}$$

$$0.2B - 2.4A = \begin{bmatrix} 1 & \frac{2}{5} & 0 \\ -1 & \frac{3}{5} & -\frac{4}{5} \\ -\frac{4}{5} & \frac{2}{5} & -\frac{4}{5} \end{bmatrix} + \begin{bmatrix} -\frac{12}{5} & \frac{24}{5} & -12 \\ -\frac{48}{5} & -\frac{48}{5} & -\frac{96}{5} \\ \frac{36}{5} & -\frac{12}{5} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{7}{5} & \frac{26}{5} & -12 \\ -\frac{47}{5} & -9 & -20 \\ \frac{32}{5} & -2 & -\frac{4}{5} \end{bmatrix}$$

$$15. (u+v)-w = \begin{bmatrix} 1.2 \\ 0 \\ -2.5 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} - \begin{bmatrix} -4 \\ -10 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 7.2 \\ 9 \\ -7.5 \end{bmatrix}$$

$$u + (v-w) = \begin{bmatrix} 7.2 \\ 9 \\ -7.5 \end{bmatrix}$$

Crow, $0E + u - v$ 는 크기가 다르기 때문에
변하지

7.2

4. $A^T = -A$ 이고 대각선 방향 요소가 0이면 스칼라를 못한다.

따라서 4x4 matrix에서 대각요소를 제외한 2개의 원소가 0일 수 있다.

$$1/2. \quad A = \begin{bmatrix} 2 & -1 & 3 \\ -2 & 1 & 4 \\ 1 & 2 & -2 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 2 & -1 & 3 \\ -2 & 1 & 4 \\ 1 & 2 & -2 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 \\ -1 & 1 & 2 \\ 3 & 4 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+1+9 & -4-1+12 & 2-2-6 \\ -4-1+12 & 4+1+16 & -2+2-8 \\ 2-2-6 & -2+2-8 & 1+4+4 \end{bmatrix}$$

$$= \begin{bmatrix} 14 & 7 & -6 \\ 7 & 21 & -8 \\ -6 & -8 & 9 \end{bmatrix}$$

$$A^* = \begin{bmatrix} 2 & -1 & 3 \\ -2 & 1 & 4 \\ 1 & 2 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \\ -2 & 1 & 4 \\ 1 & 2 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+2+3 & -2-1+6 & 6-4-6 \\ -4-2+4 & 2+1+8 & -6+4-8 \\ 2-4-2 & -1+2-4 & 3+8+4 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 3 & -4 \\ -2 & 11 & -10 \\ -4 & -3 & 15 \end{bmatrix}$$

$$BB^T = \begin{bmatrix} -1 & 3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & -3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1+9 & 3+3 & 0 \\ 3+3 & 9+1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 6 & 0 \\ 6 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} -1 & 3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1+9 & -3+3 & 0 \\ 3-3 & -9+3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$20. \quad A = \begin{bmatrix} 2 & -1 & 3 \\ -2 & 1 & 4 \\ 1 & 2 & -2 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 1 \\ -2 & 2 \\ 2 & 0 \end{bmatrix} \quad a = [-1 \ -2 \ 0] \quad b = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

$$b^T A b = \begin{bmatrix} 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \\ -2 & 1 & 4 \\ 1 & 2 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 6 + 1 + 3 \\ -6 - 1 + 4 \\ 3 - 2 - 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ -3 \\ -1 \end{bmatrix}$$

$$= 30 + 3 - 1 = 32$$

$$a B a^T$$

$$= [-1 \ -2 \ 0] \begin{bmatrix} -1 & 3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$$

$$= [-1 \ -2 \ 0] \begin{bmatrix} 1 - 4 \\ 3 - 2 \\ 0 \end{bmatrix}$$

$$= 5 - 2 = 3$$

$$a C C^T = [-1 \ -2 \ 0] \begin{bmatrix} 1 & 1 \\ -2 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

$$= [-1 \ -2 \ 0] \begin{bmatrix} 2 & 0 & 2 \\ 0 & 8 & -4 \\ -2 & -4 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & 0 \\ 0 & -16 & 0 \\ -2 & 8 & 0 \end{bmatrix}$$

$$C^T b a = \begin{bmatrix} 1 & -2 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} [-1 \ -2 \ 0]$$

$$= \begin{bmatrix} 1 & -2 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} -3 & -6 & 0 \\ 1 & 2 & 0 \\ -1 & -2 & 0 \end{bmatrix}$$

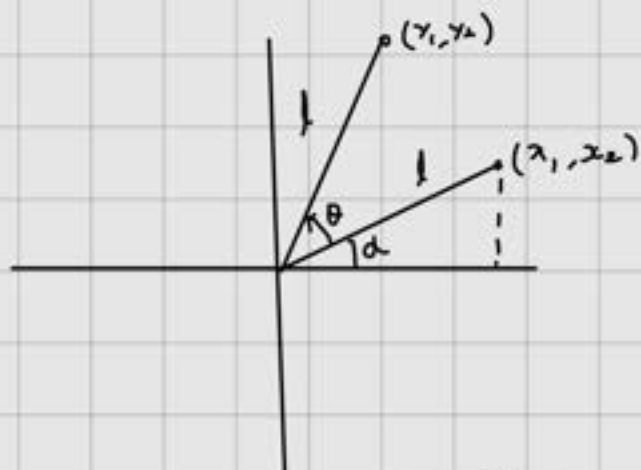
$$= \begin{bmatrix} -3 + 2 - 2 & -6 - 4 + 4 & 0 \\ -3 + 2 & -6 + 4 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -7 & -14 & 0 \\ -1 & -2 & 0 \end{bmatrix}$$

30. a)

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



$$y_1 = l(\cos(\alpha + \theta)) = l(\cos \alpha \cos \theta - \sin \alpha \sin \theta) = x_1 \cos \theta - x_2 \sin \theta$$

$$y_2 = l(\sin(\alpha + \theta)) = l(\sin \alpha \cos \theta + \cos \alpha \sin \theta) = x_2 \cos \theta + x_1 \sin \theta$$

$$x_1 = l \cos \alpha \quad \nearrow$$

$$x_2 = l \sin \alpha$$

$$\therefore \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

17.3

1. $-3x + 8y = 5$

$8x - 12y = -11$

$$\begin{bmatrix} -3 & 8 \\ 8 & -12 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -11 \end{bmatrix}$$

Augmented Matrix

$$\begin{bmatrix} -3 & 8 & 5 \\ 8 & -12 & -11 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -3 & 8 & 5 \\ 0 & \frac{4}{3} & \frac{40}{3} - 11 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 8 & 5 \\ 0 & \frac{28}{3} & \frac{7}{3} \end{bmatrix}$$

back substitution

$$\therefore \frac{28}{3}y = \frac{7}{3}$$

$$y = \frac{7}{28} = 0.25$$

$$-3x + 8y = 5$$

$$x = -1$$

B. $10x + 4y - 2z = 14$

$$-3w - 15x + y + 2z = 0$$

$$w + x + y = 6$$

$$8w - 5x + 5y - 10z = 26$$

$$\begin{bmatrix} 0 & 10 & 4 & -2 \\ -3 & -15 & 1 & 2 \\ 1 & 1 & 1 & 0 \\ 8 & -5 & 5 & -10 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ 0 \\ 6 \\ 26 \end{bmatrix}$$

Augmented Matrix

$$\begin{bmatrix} 0 & 10 & 4 & -2 & 14 \\ -3 & -15 & 1 & 2 & 0 \\ 1 & 1 & 1 & 0 & 6 \\ 8 & -5 & 5 & -10 & 26 \end{bmatrix}$$

->

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 6 \\ -3 & -15 & 1 & 2 & 0 \\ 0 & 10 & 4 & -2 & 14 \\ 8 & -5 & 5 & -10 & 26 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 6 \\ 0 & -12 & 4 & 2 & 18 \\ 0 & 10 & 4 & -2 & 14 \\ 0 & -13 & -3 & -10 & -22 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 6 \\ 0 & -12 & 4 & 2 & 18 \\ 0 & 0 & \frac{22}{3} & -\frac{1}{3} & 24 \\ 0 & 0 & -\frac{22}{3} & -\frac{19}{6} & -\frac{83}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 6 \\ 0 & -12 & 4 & 2 & 18 \\ 0 & 0 & \frac{22}{3} & -\frac{1}{3} & 24 \\ 0 & 0 & 0 & -\frac{25}{2} & -\frac{35}{2} \end{bmatrix}$$

$$\therefore z = 1$$

$$\frac{22}{3}y - \frac{1}{3}z = 24$$

$$y = 4$$

$$-12x + 4y + 2z = 18$$

$$x = 0$$

$$w + x + y = 6$$

$$\therefore w = -4$$

9. $-2y - 2z = 8$

$$3x + 4y - 5z = 8$$

$$\begin{bmatrix} 3 & 4 & -5 \\ 0 & -2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 4 & -5 & 8 \\ 0 & -2 & -2 & 8 \end{bmatrix}$$

$$\text{rank } A = 2$$

$$\text{rank}(\text{Augmented } A) = 2$$

$$\text{rank}(\text{matrix coeffs}) = 3 \text{ 이므로}$$

$$\therefore \text{해가 무한히 많다.}$$

7.4

$$1) \begin{bmatrix} -2 & 4 & -6 \\ 1 & -2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 4 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank } A = 1 \quad \text{basis: } [-2 \ 4 \ -6]$$

$$3) \begin{bmatrix} 0 & 0 & 5 \\ 3 & 5 & 0 \\ 5 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 5 \\ 0 & 5 & 0 \\ 5 & 0 & 0 \end{bmatrix}$$

$$\text{rank } A = 3 \quad \text{basis: } \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

$$5) \begin{bmatrix} 2 & -2 & 1 \\ 0 & 4 & 8 \\ 2 & 0 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 & 1 \\ 0 & 4 & 8 \\ 0 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -2 & 1 \\ 0 & 4 & 8 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\therefore \text{rank } A = 3 \quad \therefore \text{basis: } \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \\ -1 \end{bmatrix}$$

$$7) \begin{bmatrix} -6 & 0 & -3 & 0 \\ 0 & -1 & 0 & 5 \\ 2 & 0 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & 0 & -3 & 0 \\ 0 & -1 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A) = 2 \quad \text{basis } [6 \ 0 \ -3 \ 0] \\ [0 \ -1 \ 0 \ 5]$$

$$9) \begin{bmatrix} 5 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 5 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & \frac{4}{3} & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 5 & 0 & 1 & 0 \\ 0 & 1 & \frac{4}{3} & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A) = 3$$

$$\text{basis: } [5 \ 0 \ 1 \ 0] \\ [0 \ 1 \ \frac{4}{3} \ 1] \\ [0 \ 0 \ -1 \ 0]$$

$$13) \text{rank} \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right) = \text{rank} \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right) = 1$$

2.7.2020

$$\text{rank} \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right) \neq \text{rank} \left(\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right)$$

$$1 \neq 0$$

$$K_1. \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore$ independent

$$21. \begin{bmatrix} 2 & 0 & 0 & 7 \\ 2 & 0 & 0 & 8 \\ 2 & 0 & 0 & 9 \\ 2 & 0 & 1 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 0 & 0 & 7 \\ 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 0 & 0 & 7 \\ 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ dependent

$$27. \mathbb{R}^3 \quad v_2 - v_1 + 4v_3 = 0$$

원점을 지나므로 vector space.

$$\text{평면을 형성하므로 } \dim(v_2 - v_1 + 4v_3 = 0) = 2$$

$$\text{basis: } \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ \frac{1}{4} \end{bmatrix}$$

$$32. \mathbb{R}^3 \quad 3v_1 - 2v_2 + v_3 = 0$$

$$4v_1 + 5v_2 = 0$$

원점을 지나므로 vector space

$$\dim = 1$$

$$\begin{bmatrix} 3 & -2 & 1 \\ 4 & 5 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 3 & -2 & 1 \\ 0 & \frac{23}{3} & -\frac{4}{3} \end{bmatrix}$$

$$\therefore \frac{23}{3}v_2 - \frac{4}{3}v_3 = 0$$

$$v_3 = \frac{23}{4}v_2$$

$$3v_1 - 2v_2 + v_3 = 0$$

$$\frac{15}{4}v_2 = -3v_1$$

$$-\frac{5}{4}v_2 = v_1$$

$$\therefore v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -\frac{5}{4}v_2 \\ v_2 \\ \frac{23}{4}v_2 \end{bmatrix}$$

$$\therefore \text{basis: } \begin{bmatrix} 5 \\ -4 \\ 23 \end{bmatrix}$$

7.7

$$5. \begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} ka & c \\ kb & d \end{vmatrix} = kad - kbc = k(ad - bc)$$

k'ul

$$8. \begin{vmatrix} 0.4 & 0.6 \\ 1.5 & -0.5 \end{vmatrix}$$

$$= -0.2 - 0.9$$

$$= -1.1$$

$$11. \begin{vmatrix} 6 & -1 & 8 \\ 0 & -2 & 9 \\ 0 & 0 & -4 \end{vmatrix}$$

$$= 6 \times 8 - 0 \times \sim + 0 \times \sim$$

$$= 48$$

14. Matlab

```
>> A = [4 7 0 0; 2 8 0 0; 0 0 1 5; 0 0 -2 2];
>> det(A)

ans =

    216
```

$$21. \quad 2x - y = 5.15$$

$$3x + 4y = 6.15$$

$$\begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5.15 \\ 6.15 \end{bmatrix}$$

$$x = \frac{\begin{vmatrix} 5.15 & -1 \\ 6.15 & 4 \end{vmatrix}}{\begin{vmatrix} 2 & -1 \\ 3 & 4 \end{vmatrix}} = \frac{52.5}{21} = 2.5$$

$$y = \frac{\begin{vmatrix} 2 & 5.15 \\ 3 & 6.15 \end{vmatrix}}{\begin{vmatrix} 2 & -1 \\ 3 & 4 \end{vmatrix}} = \frac{-3.15}{21} = -0.15$$

$$22. \quad 3x + 2y = 7$$

$$2y - 3x = 2$$

$$-x + 4y = 1$$

$$\begin{bmatrix} 3 & 2 & 0 \\ 0 & 2 & -3 \\ -1 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \\ 1 \end{bmatrix}$$

$$D = \begin{vmatrix} 3 & 2 & 0 \\ 0 & 2 & -3 \\ -1 & 0 & 4 \end{vmatrix} = 24 + 6 + 0 = 30$$

$$D_1 = \begin{vmatrix} -1 & 2 & 0 \\ 2 & 2 & -3 \\ 1 & 0 & 4 \end{vmatrix} = -8 - 6 - 16 = -30$$

$$D_2 = \begin{vmatrix} 3 & -1 & 0 \\ 0 & 2 & -3 \\ -1 & 1 & 4 \end{vmatrix} = 24 - 3 + 1 = 30$$

$$D_3 = \begin{vmatrix} 3 & 2 & -1 \\ 0 & 2 & 2 \\ -1 & 0 & 1 \end{vmatrix} = 6 - 4 - 2 = 0$$

$$\therefore x = \frac{D_1}{D} = -1$$

$$y = \frac{D_2}{D} = 1$$

$$z = \frac{D_3}{D} = 0$$

$$9. \begin{vmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{vmatrix}$$

$$= \cos^2 \theta + \sin^2 \theta$$

$$= 1$$

$$11.8.3 \begin{bmatrix} 0 & -0.2 & 0.75 \\ 0.4 & 1 & 2 \\ 0 & 0 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -0.2 & 0.75 & 1 & 0 & 0 \\ 0.4 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 0.4 & 1 & 2 & 0 & 1 & 0 \\ 0 & -0.2 & 0.75 & 1 & 0 & 0 \\ 0 & 0 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 0.4 & 1 & 0 & 0 & 1 & -\frac{1}{4} \\ 0 & -0.2 & 0 & 1 & 0 & -\frac{3}{32} \\ 0 & 0 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 0.4 & 0 & 0 & 5 & 1 & -\frac{23}{32} \\ 0 & -0.2 & 0 & 1 & 0 & -\frac{3}{32} \\ 0 & 0 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & \frac{25}{2} & 5 & \frac{115}{64} \\ 0 & 1 & 0 & -5 & 0 & \frac{15}{32} \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{8} \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{25}{2} & 5 & \frac{115}{64} \\ -5 & 0 & \frac{15}{32} \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$$

$$5. \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ -2 & 1 & 0 & 0 & 1 & 0 \\ 5 & -4 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & -4 & 1 & -5 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 3 & 4 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix}$$

$$10. \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} & 1 & 0 & 0 \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} & 0 & 1 & 0 \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} & 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 1 & 2 & 3 & 0 & 0 \\ -2 & 2 & 1 & 0 & 3 & 6 \\ 1 & 2 & -2 & 0 & 0 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 1 & 2 & 3 & 0 & 0 \\ 6 & 3 & 3 & 3 & 3 & 6 \\ 0 & 1.5 & -3 & -1.5 & 0 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 1 & 2 & 3 & 0 & 0 \\ 6 & 3 & 3 & 3 & 3 & 0 \\ 0 & 0 & -\frac{9}{2} & -3 & -\frac{3}{2} & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 1 & 0 & \frac{5}{3} & -\frac{2}{3} & \frac{4}{3} \\ 6 & 3 & 0 & 1 & 2 & 2 \\ 0 & 0 & -\frac{9}{2} & -3 & -\frac{3}{2} & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 0 & 0 & \frac{4}{3} & -\frac{4}{3} & \frac{2}{3} \\ 6 & 1 & 0 & \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 1 & \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} \\ 6 & 1 & 0 & \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 1 & \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix}$$

```
>> clear
>> A = [2/3 1/3 2/3; -2/3 2/3 1/3; 1/3 2/3 -2/3];
>> inv(A)
```

ans =

```
0.6667    -0.6667    0.3333
0.3333    0.6667    0.6667
0.6667    0.3333   -0.6667
```

11.

$$(A^*)^{-1} = (A^{-1})^*$$

$$A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -5 & -\frac{3}{2} \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -5 & -\frac{3}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -5 & -\frac{3}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{4} + \frac{5}{4} & -\frac{1}{8} + \frac{3}{8} \\ -\frac{5}{2} + \frac{15}{2} & \frac{5}{4} + \frac{9}{4} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{2} & \frac{1}{4} \\ 5 & \frac{13}{2} \end{bmatrix}$$

$$[A^2: I]$$

$$\begin{bmatrix} \frac{3}{2} & \frac{1}{4} & 1 & 0 \\ 5 & \frac{13}{2} & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 6 & 1 & 4 & 0 \\ 10 & 13 & 0 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 6 & 0 & \frac{21}{4} & -\frac{6}{16} \\ 0 & \frac{16}{3} & -\frac{26}{3} & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & \frac{7}{8} & -\frac{1}{16} \\ 0 & 1 & -\frac{5}{4} & \frac{3}{8} \end{bmatrix}$$

$$\therefore (A^2)^{-1} = \begin{bmatrix} \frac{7}{8} & -\frac{1}{16} \\ -\frac{5}{4} & \frac{3}{8} \end{bmatrix} \cdot (A^*)^2$$

$$[A: I]$$

$$\begin{bmatrix} \frac{1}{2} & -\frac{1}{4} & 1 & 0 \\ -5 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & -1 & 4 & 0 \\ -10 & -3 & 0 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 0 & \frac{3}{2} & -\frac{1}{4} \\ 0 & -8 & 20 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & \frac{3}{4} & -\frac{1}{8} \\ 0 & 1 & -\frac{5}{2} & -\frac{1}{4} \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{3}{4} & -\frac{1}{8} \\ -\frac{5}{2} & -\frac{1}{4} \end{bmatrix}$$

$$(A^{-1})^2 = \begin{bmatrix} \frac{3}{4} & -\frac{1}{8} \\ -\frac{5}{2} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} \frac{3}{4} & -\frac{1}{8} \\ -\frac{5}{2} & -\frac{1}{4} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{9}{16} + \frac{5}{16} & -\frac{3}{32} + \frac{1}{32} \\ -\frac{15}{8} + \frac{5}{8} & \frac{5}{16} + \frac{1}{16} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{7}{8} & -\frac{1}{16} \\ -\frac{5}{4} & \frac{3}{8} \end{bmatrix}$$

$$12. (A^2)^{-1} = (A^{-1})^2$$

$$\text{let } A^2 = B$$

$$B B^{-1} = I$$

$$A^2 B^{-1} = I$$

$$B^{-1} = (A^2)^{-1}$$

$$(A^2)^{-1} = (A^{-1})^2$$

$$14. (A^T)^{-1} = (A^{-1})^T$$

$$\text{let } A^T = B, A = B^T$$

$$B B^{-1} = I$$

$$(B B^{-1})^T = I$$

$$B^T (B^{-1})^T = I$$

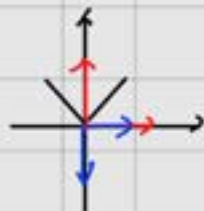
$$(B^{-1})^T = A^{-1}$$

$$(B^{-1})^T = (B^T)^{-1}$$

11.9

1. $\mathbb{R}^2$ 에서

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



3. In $\mathbb{R}^3$

$$-2v_1 + v_2 + 4v_3 = 0 \quad - \text{평면}$$

$$v_1 - v_2 - v_3 = 0 \quad - \text{평면}$$

$\therefore$ 직선을 얻는다.

$$\begin{bmatrix} -2 & 1 & 4 \\ 1 & -1 & -1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -2 & 1 & 4 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix}$$

$$-\frac{1}{2}v_2 + v_3 = 0$$

$$v_2 = 2v_3$$

$$-2v_1 + v_2 + 4v_3 = 0$$

$$-2v_1 + 6v_3 = 0$$

$$v_1 = 3v_3$$

$$\therefore \text{basis: } \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

15. $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \quad \|v\| = \sqrt{2^2 + (-1)^2 + 3^2}$
 $= \sqrt{14}$

17. $[1 \ -1 \ 1 \ 1 \ -1 \ -1 \ -1 \ 1]^T$

$$\|v\| = \sqrt{1+1+1+1+(-1)+(-1)+(-1)+1}$$

$$= 2\sqrt{2}$$

21. use Inner product

$$\begin{bmatrix} 1 \\ 0.5 \\ -4 \\ k \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 0 \\ \frac{1}{2} \end{bmatrix} = -1 + 0.5 + 0 + \frac{k}{2} = 0$$

$$\therefore k = 1$$

22.

$$\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$2x + z = 0 \quad \text{문 만족하는 Vector}$$

$$\therefore \begin{bmatrix} x \\ y \\ -2x \end{bmatrix}$$

$(0,0,0)$ 을 제외

Vector space.

Review 2 3 11 14 15 16 17 22 23

217

2. $A: 100 \times 100$
 $B: 100 \times 50$ Newton

AB : 크기 알

A^2 : ok $(100 \times 100)(100 \times 100)$

B^2 : X $(100 \times 50)(100 \times 50)$

AB : ok $(100 \times 100)(100 \times 50)$

BA : X $(100 \times 50)(100 \times 100)$

AA^T : ok $(100 \times 100)(100 \times 100)$

$B^T A$: ok $(50 \times 100)(100 \times 100)$

$B^T B$: ok $(50 \times 100)(100 \times 50)$

BB^T : ok $(100 \times 50)(50 \times 100)$

$B^T AB$: $(50 \times 100)(100 \times 100)(100 \times 50)$
 $(50 \times 100)(100 \times 50)$

(50×50) ok

3. $AX = b$ 가라

$\text{rank}(A) = \text{rank}(\text{Augmented } A)$ ok

Sol 풀기

$\begin{cases} x+y=3 \\ x+2=2 \end{cases}$ No Sol.

$\begin{cases} x+y=3 \\ x+2y=4 \end{cases}$ unique Sol.

$\begin{cases} x+y+2=3 \\ 2x+2y+22=6 \end{cases}$ 무한히 많은 Sol

11. $A = \begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 6 & 2 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}$

$u = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ $v = \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$

$$AB = \begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 6 & 2 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 4 & -2 & 2 & -1 & 4 \\ 0 & 2 & 1 & 1 & -2 \\ -1 & -4 & -1 & -2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 5 \\ 0 & 3 & -1 \\ -1 & -5 & -1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 6 & 2 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 1 & 2 & -1 \\ -2 & 1 & 2 & -1 & 1 \\ 1 & 1 & -2 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 1 \\ 0 & -2 & 3 \\ 5 & -2 & 3 \end{bmatrix}$$

14. $u^T v = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix} = -2 + 1 = -1$

$$u v^T = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} -2 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & -1 \\ 0 & 0 & 0 \\ 2 & -1 & 1 \end{bmatrix}$$

15. $u^T A u = u^T \begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$
 $= \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix}$

$$= 4 + 1 = 5$$

$$r^T B r = \begin{bmatrix} -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 6 & 2 \\ -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 2 & +1 \\ 1 & +2 \end{bmatrix}$$

$$= -2 + 3 - 3 = -2$$

16. A^{-1}

$$[A|I] = \begin{bmatrix} 2 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ -2 & 1 & -1 & 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 1 & -2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & 2 & -\frac{1}{2} & 1 & 0 \\ 0 & 2 & -3 & 1 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 1 & -2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & 2 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 5 & -1 & 4 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 1 & 0 & \frac{7}{5} & \frac{8}{5} & \frac{2}{5} \\ 0 & -\frac{1}{2} & 0 & -\frac{1}{10} & -\frac{3}{5} & -\frac{2}{5} \\ 0 & 0 & 5 & -1 & 4 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 0 & 0 & \frac{2}{5} & \frac{2}{5} & -\frac{2}{5} \\ 0 & -\frac{1}{2} & 0 & -\frac{1}{10} & -\frac{3}{5} & -\frac{2}{5} \\ 0 & 0 & 5 & -1 & 4 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & -\frac{1}{5} \\ 0 & 1 & 0 & \frac{1}{5} & \frac{6}{5} & \frac{4}{5} \\ 0 & 0 & 1 & -\frac{1}{5} & \frac{4}{5} & \frac{1}{5} \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{6}{5} & \frac{4}{5} \\ -\frac{1}{5} & \frac{4}{5} & \frac{1}{5} \end{bmatrix}$$

B^{-1}

$$[B|I] = \begin{bmatrix} 6 & 2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -1 & 0 & 1 & 0 & 0 \\ 6 & 2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -1 & 0 & 1 & 0 & 0 \\ 6 & 2 & 1 & 0 & 0 \\ 0 & 0 & -\frac{5}{2} & -\frac{1}{2} & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -1 & 0 & \frac{1}{5} & -\frac{2}{5} \\ 6 & 2 & \frac{4}{5} & \frac{2}{5} \\ 0 & 0 & -\frac{5}{2} & -\frac{1}{2} & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{5} & \frac{2}{5} \\ 6 & 1 & \frac{2}{5} & \frac{1}{5} \\ 0 & 0 & 1 & \frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$\therefore B^{-1} = \begin{bmatrix} -\frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$17. \det(A) = \begin{vmatrix} 2 & 1 & -2 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{vmatrix} = -2 \cdot 2 + 1 \cdot 2 = -5$$

$$A^2 = \begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 4+1+4 & 2-2 & -4+1+2 \\ 2-2 & 1+1 & -2-1 \\ -4+1+2 & -2-1 & 4+1+1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 & -1 \\ 0 & 2 & -3 \\ -1 & -3 & 6 \end{bmatrix}$$

$$\det(A^2) = \begin{vmatrix} 9 & 0 & -1 \\ 0 & 2 & -3 \\ -1 & -3 & 6 \end{vmatrix} = 108 - 2 - 81 = 25$$

$$\det(A) \det(A) = 25$$

$$\det(B) = \begin{vmatrix} 6 & 2 \\ -1 & 0 \\ 0 & 1 \end{vmatrix} = -1 \cdot 4 = -5$$

$$22. \quad 5x - 3y + z = 7$$

$$2x + 3y - z = 0$$

$$8x + 1y - 3z = 2$$

$$\begin{bmatrix} 5 & -3 & 1 \\ 2 & 3 & -1 \\ 8 & 1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \\ 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 5 & -3 & 1 & 7 \\ 0 & \frac{21}{5} & -\frac{1}{5} & -\frac{14}{5} \\ 0 & \frac{41}{5} & -\frac{28}{5} & -\frac{18}{5} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 5 & -3 & 1 & 7 \\ 0 & \frac{21}{5} & -\frac{1}{5} & -\frac{14}{5} \\ 0 & 0 & -\frac{28}{3} & -\frac{40}{3} \end{bmatrix}$$

$$\therefore -\frac{28}{3}z = -\frac{40}{3}$$

$$z = 2$$

$$\frac{21}{5}y - \frac{1}{5}z = -\frac{14}{5}$$

$$y = 0$$

$$5x - 3y + z = 7$$

$$x = 1$$

$$23. \quad 9x + 3y - 6z = 60$$

$$2x - 4y + 8z = 4$$

$$\begin{bmatrix} 9 & 3 & -6 \\ 2 & -4 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 60 \\ 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 9 & 3 & -6 & 60 \\ 2 & -4 & 8 & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 9 & 3 & -6 & 60 \\ 0 & -\frac{14}{3} & \frac{24}{1} & -\frac{84}{3} \end{bmatrix} \quad -\frac{1}{9}$$

$$-\frac{14}{3}y + \frac{24}{1}z = -\frac{84}{3}$$

$$y + 2z = 2 \quad \text{of } x, y, z$$

$$x = 6$$

$$27. \quad -x + 2y = 0$$

$$4x - 2y = -3$$

$$3x - 5y = -0.5$$

$$\text{rank}(A) = 2$$

$$4 > 0 \text{ and } 5 : \text{ unique sol.}$$

$$\rightarrow \begin{bmatrix} -1 & 2 & 0 \\ 4 & -2 & -3 \\ 3 & -5 & -0.5 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 0 \\ 0 & 6 & -3 \\ 0 & 1 & -0.5 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -1 & 2 & 0 \\ 0 & 6 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore 6y = -3$$

$$y = -\frac{1}{2}$$

$$-x + 2y = 0$$

$$x = -1$$

$$\text{rank}(A) = 2$$

8.

$$1. \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & 3 \end{bmatrix}$$

$$AX = \lambda X$$

$$(A - \lambda I)X = 0$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} \frac{3}{2} - \lambda & 0 \\ 0 & 3 - \lambda \end{vmatrix} = 0$$

$$(\lambda - 3)(\lambda - \frac{3}{2}) = 0$$

$$\lambda = \frac{3}{2}, 3$$

$$\lambda = \frac{3}{2}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & \frac{3}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\frac{3}{2}x_2 = 0$$

$$x_2 = 0$$

$$\therefore \text{eigenvektor} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} -\frac{3}{2} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-\frac{3}{2}x_1 = 0$$

$$x_1 = 0$$

$$\therefore \text{eigenvektor} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$3. \begin{bmatrix} 3 & -2 \\ 4 & -6 \end{bmatrix} X = \lambda X$$

$$\begin{vmatrix} 3 - \lambda & -2 \\ 4 & -6 - \lambda \end{vmatrix} = 0$$

$$-18 + 3\lambda + \lambda^2 + 18 = 0$$

$$\lambda(\lambda + 3) = 0$$

$$\lambda = -3, 0$$

$$\lambda = -3$$

$$\begin{bmatrix} 6 & -2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\therefore \text{eigenvektor} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\lambda = 0$$

$$\begin{bmatrix} 3 & -2 \\ 4 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\therefore \text{eigenvektor} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$6. \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} X = \lambda X$$

$$\begin{vmatrix} 1 - \lambda & 2 \\ 0 & 3 - \lambda \end{vmatrix} = 0$$

$$3 - 4\lambda + \lambda^2 = 0$$

$$\lambda = 1, 3$$

$$\lambda = 1$$

$$\begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$2x_2 = 0$$

$$x_2 = 0$$

$$\therefore \text{eigenvektor} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 = x_2$$

$$\therefore \text{eigenvektor} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$12. \begin{bmatrix} 3 & 5 & 3 \\ 0 & 4 & 6 \\ 6 & 6 & 1 \end{bmatrix} X = \lambda X$$

$$\begin{vmatrix} 3 - \lambda & 5 & 3 \\ 0 & 4 - \lambda & 6 \\ 6 & 6 & 1 - \lambda \end{vmatrix} = 0$$

$$(3 - \lambda)(4 - \lambda)(1 - \lambda) = 0$$

$$\lambda = 1, 3, 4$$

$$\lambda = 1$$

$$\begin{bmatrix} 2 & 5 & 3 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$2x_1 + 5x_2 + 3x_3 = 0$$

$$3x_2 + 6x_3 = 0$$

$$x_2 = -2x_3$$

$$\therefore x_1 = \frac{1}{2}x_3$$

$$\therefore \text{eigenvektor} \begin{bmatrix} \frac{1}{2} \\ -2 \\ 1 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 6 \\ 6 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$-2x_3 = 0$$

$$x_3 = 0$$

$$x_2 + 6x_3 = 0$$

$$x_2 = 0$$

$$\therefore \text{eigenvektor} : \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 4$$

$$\begin{bmatrix} -1 & 5 & 3 \\ 0 & 0 & 6 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$x_3 = 0$$

$$-x_1 + 5x_2 + 3x_3 = 0$$

$$\therefore x_1 = 5x_2$$

$$\therefore \text{eigenvektor} : \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$$

$$15. A = \begin{bmatrix} -1 & 0 & 12 & 0 \\ 0 & -1 & 0 & 12 \\ 0 & 0 & -1 & -4 \\ 0 & 0 & -4 & -1 \end{bmatrix}$$

$$Ax = \lambda x$$

$$\begin{vmatrix} -1-\lambda & 0 & 12 & 0 \\ 0 & -1-\lambda & 0 & 12 \\ 0 & 0 & -1-\lambda & -4 \\ 0 & 0 & -4 & -1-\lambda \end{vmatrix} = 0$$

```
>> A = [-1 0 12 0; 0 -1 0 12; 0 0 -1 -4; 0 0 -4 -1]
```

```
A =
```

```

-1    0    12    0
 0    -1    0    12
 0     0    -1   -4
 0     0    -4   -1

```

```
>> eig(A)
```

```
ans =
```

```

-1
-1
 3
-5

```

$$\lambda = -5, -1(\text{2}), 3$$

$$\lambda = -5 \quad \begin{bmatrix} 4 & 0 & 12 & 0 \\ 0 & 4 & 0 & 12 \\ 0 & 0 & 4 & -4 \\ 0 & 0 & -4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$-4x_3 + 4x_4 \rightarrow x_3 = x_4$$

$$4x_2 + 12x_4 \rightarrow x_2 = -3x_4$$

$$4x_1 + 12x_3 \rightarrow x_1 = -3x_3$$

$$\therefore \text{Eigenvektor} \begin{bmatrix} -3 \\ -3 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda = -1(\text{2})$$

$$\begin{bmatrix} 0 & 0 & 12 & 0 \\ 0 & 0 & 0 & 12 \\ 0 & 0 & 0 & -4 \\ 0 & 0 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$x_3 = x_4 = 0$$

$$\therefore \text{Eigenvektor} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} -4 & 0 & 12 & 0 \\ 0 & -4 & 0 & 12 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & -4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

$$-4x_3 - 4x_4 = 0$$

$$x_3 = -x_4$$

$$-4x_2 + 12x_4 = 0$$

$$x_2 = 3x_4$$

$$-4x_1 + 12x_3 = 0$$

$$x_1 = -3x_3 = 3x_4$$

$$\therefore \text{Eigenvektor} \begin{bmatrix} 3 \\ 3 \\ -1 \\ 1 \end{bmatrix}$$

17.



$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} x = \lambda x$$

$$\begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda = -i, +i$$

$$\lambda = -i$$

$$\begin{bmatrix} i & 1 \\ 1 & i \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$\lambda = i$$

$$\begin{bmatrix} -i & 1 \\ 1 & -i \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

90° 회전하여 2 방향이 바뀌는 Vector는 $\mathbb{R}^2$ 에 없다.

24.

$$Ax = \lambda x$$

$$A^T Ax = A^T \lambda x$$

$$x = A^{-1} \lambda x$$

$$\frac{1}{\lambda} x = A^{-1} x$$

$$\therefore \lambda \neq 0$$

8.2

$$1. \begin{bmatrix} 0.5 & 1.5 \\ 1.5 & 0.5 \end{bmatrix}$$

$$\text{Det}(A - \lambda I) = 0$$

$$(0.5 - \lambda)^2 - 1.5^2 = 0$$

$$0.5 - \lambda - 1.5 = 0 \quad \text{or} \quad 0.5 - \lambda + 1.5 = 0$$

$$\lambda = 2$$

$$\lambda = -1$$

$\lambda = -1$ 일 때

$\lambda = 2$ 일 때

$$\begin{bmatrix} 1.5 & 1.5 \\ 1.5 & 1.5 \end{bmatrix} x = 0 \quad \begin{bmatrix} -1.5 & 1.5 \\ 1.5 & -1.5 \end{bmatrix} x = 0$$

$$\therefore x = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \triangle$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \triangle$$

$\therefore$ principal dir

-45°

$\therefore$ principal dir

45°

$$16. A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda - a & -b \\ -c & \lambda - d \end{vmatrix} = 0$$

$$(\lambda - a)(\lambda - d) - bc = 0$$

$$\lambda^2 - (a + d)\lambda + ad - bc = 0$$

$$\lambda = \frac{a + d \pm \sqrt{(a + d)^2 - 4(ad - bc)}}{2}$$

$$\lambda_1 + \lambda_2 = a + d$$

주대값들의 합 $a + d$) Same

8.3

$$A^T = A^{-1} \text{ (orthogonal)} \quad A^T = -A \text{ (skew symmetric)} \quad A = A^T \text{ (symmetric)}$$

$$1. A = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix} \therefore \text{orthogonal} \\ A^T = A^{-1} \\ = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

3 -4

4 3

$$4. \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\text{직접 확인 가능} \quad \text{orthogonal} \quad A^T = A^{-1}$$

$$7. A = \begin{bmatrix} 0 & -1 & -\frac{1}{2} \\ 1 & 0 & 1 \\ \frac{1}{2} & -1 & 0 \end{bmatrix}$$

$$\det(A) = 1 \times \begin{vmatrix} 1 & 1 \\ \frac{1}{2} & 0 \end{vmatrix} - \frac{1}{2} \begin{vmatrix} 1 & 0 \\ \frac{1}{2} & -1 \end{vmatrix} \\ = -\frac{1}{2} + \frac{1}{2} = 0$$

 $\therefore A^{-1}$ 존재 X

$$\therefore \text{skew-symmetric} \quad A^T = -A$$

$$8. A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\det(A) = (\cos^2 \theta + \sin^2 \theta) = 1$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$

$$A^{-1} = A^T \therefore \text{orthogonal}$$

$$10. A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

[A:I]

$$\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} & 1 & 0 & 0 \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} & 0 & 1 & 0 \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} & 0 & 0 & 1 \end{bmatrix}$$

 $\rightarrow$

$$\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} & 1 & 0 & 0 \\ 0 & -1 & 2 & -2 & 1 & 0 \\ 0 & 2 & -1 & 2 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} & 1 & 0 & 0 \\ 0 & -1 & 2 & -2 & 1 & 0 \\ 0 & 0 & 3 & -2 & 2 & 1 \end{bmatrix}$$

 $\rightarrow$

$$\begin{bmatrix} 1 & 2 & 0 & \frac{5}{3} & \frac{4}{3} & \frac{2}{3} \\ 0 & -1 & 0 & -\frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

 $\rightarrow$

$$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & +1 & 0 & +\frac{2}{3} & +\frac{1}{3} & +\frac{2}{3} \\ 0 & 0 & 1 & -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ +\frac{2}{3} & +\frac{1}{3} & +\frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

$A^T = A^{-1} \therefore$ orthogonal and symmetric

15. $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$
 eigenvalue 0 eigenvalue 0

$$A+B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\lambda^2 - 1 = 0$$

$$\lambda = -1, 1$$

$\therefore$ Answer: No.

16. $A^T = A$ symmetric

$$Ax = \alpha x, Ay = \beta y \rightarrow \text{Show that } x^T y = 0$$

$$\alpha x^T = x^T A^T = x^T A \downarrow$$

$$x^T y = 0$$

$$\alpha x^T y = x^T A y = \beta x^T y$$

different eigenvalue 이므로 등식 성립이 무리

$$\underline{x^T y = 0}$$

8.4

$$1. A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\begin{vmatrix} \lambda - 3 & -4 \\ -4 & \lambda + 3 \end{vmatrix} = 0$$

$$\lambda^2 - 9 - 16 = 0$$

$$\lambda = \pm 5$$

$$\lambda = -5$$

$$\begin{bmatrix} -8 & -4 \\ -4 & -2 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\lambda = 5$$

$$\begin{bmatrix} 2 & -4 \\ -4 & 8 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$4. A = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}$$

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda & 0 & -2 \\ 0 & \lambda - 3 & -2 \\ -1 & 0 & \lambda - 1 \end{vmatrix} = 0$$

$$\lambda(\lambda - 3)(\lambda - 1) + 2(\lambda - 3)(-1) = 0$$

$$\lambda^3 - 4\lambda^2 + 3\lambda - 2\lambda + 6 = 0$$

$$2 \mid \lambda^3 - 4\lambda^2 + \lambda + 6 = 0$$

$$\begin{array}{r|rrr} & \lambda^3 & -4\lambda^2 & \lambda + 6 \\ & 2 & -4 & 6 \end{array}$$

$$+ \begin{array}{r|rrr} & 1 & -2 & -3 \\ & -1 & 3 & \end{array}$$

$$\begin{array}{r|rrr} & 1 & -3 & 0 \end{array}$$

$$\lambda = -1, 2, 3$$

$$\lambda = -1$$

$$\begin{bmatrix} -1 & 0 & -2 \\ 0 & -4 & -2 \\ -1 & 0 & -2 \end{bmatrix} X = 0 \quad X_1 = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix}$$

$$\lambda = 2$$

$$\begin{bmatrix} 2 & 0 & -2 \\ 0 & -1 & -2 \\ -1 & 0 & 1 \end{bmatrix} X_2 = 0 \quad X_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} 3 & 0 & -2 \\ 0 & 0 & -2 \\ -1 & 0 & 2 \end{bmatrix} X_3 = 0 \quad X_3 = \begin{bmatrix} 0 \\ 1 \\ 6 \end{bmatrix}$$

$$P = \begin{bmatrix} 4 & 1 & 0 \\ 1 & -2 & 1 \\ -2 & 1 & 0 \end{bmatrix} \quad \det(P) = -2 - 4 = -6$$

$$P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & -4 \\ -3 & -6 & -9 \end{bmatrix} = -\frac{1}{6} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & -4 \\ -3 & -6 & -9 \end{bmatrix}$$

$$\therefore \hat{P} = P^{-1}AP =$$

$$-\frac{1}{6} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & -4 \\ -3 & -6 & -9 \end{bmatrix} \begin{bmatrix} 0 & 0 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 0 \\ 1 & -2 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

$$= -\frac{1}{6} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & -4 \\ -3 & -6 & -9 \end{bmatrix} \begin{bmatrix} -4 & 2 & 0 \\ -1 & -4 & 3 \\ 2 & 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\therefore \hat{P} = \text{eigenvalue}$$

$$\lambda = -1, 2, 3$$

$$\lambda I - \hat{P} = \begin{bmatrix} \lambda + 1 & 0 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 3 \end{bmatrix}$$

$$\lambda = -1 \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix} Y = 0 \quad \therefore Y = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \lambda+1 & 0 & 0 \\ 0 & \lambda-2 & 0 \\ 0 & 0 & \lambda-3 \end{bmatrix}$$

$$\lambda=2$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} y = 0 \quad y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda=3$$

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} y = 0 \quad y = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$x = Py$$

$$P^{-1}x = y \text{ is simpler.}$$

$$9. A = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$$

$$|\lambda I - A| = 0$$

$$(\lambda-2)^2 - 16 = 0$$

$$\lambda^2 - 4\lambda - 12 = 0$$

$$\lambda = -2, 6$$

$$\lambda = -2$$

$$\lambda = 6$$

$$\begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix} x = 0$$

$$\begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\therefore P^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\therefore \hat{P} = P^{-1}AP = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & 6 \end{bmatrix}$$

$$13. A = \begin{bmatrix} 6 & 0 & 0 \\ 12 & 2 & 0 \\ 21 & -6 & 4 \end{bmatrix}$$

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda-6 & 0 & 0 \\ -12 & \lambda-2 & 0 \\ -21 & 6 & \lambda-4 \end{vmatrix} = 0$$

$$(\lambda-6)(\lambda-2)(\lambda-4) = 0$$

$$\lambda = 2, 6, 4$$

$$\lambda = 2$$

$$\begin{bmatrix} -4 & 0 & 0 \\ -12 & 0 & 0 \\ -21 & 6 & -2 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 0 \\ 1 \\ 6 \end{bmatrix}$$

$$\lambda = 6$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -12 & 4 & 0 \\ -21 & 6 & -2 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$$

$$\lambda = 4$$

$$\begin{bmatrix} 3 & 0 & 0 \\ -12 & 11 & 0 \\ -21 & 6 & 0 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 0 & 1 & 0 \\ 11 & 3 & 0 \\ 6 & -1 & 1 \end{bmatrix}$$

$$P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} 3 & -1 & 0 \\ -11 & 6 & 0 \\ -25 & 6 & -1 \end{bmatrix}$$

$$\det(P) = -11$$

$$\therefore \hat{P} = P^{-1}AP$$

$$= \frac{1}{-11} \begin{bmatrix} 3 & -1 & 0 \\ -11 & 6 & 0 \\ -25 & 6 & -1 \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 12 & 2 & 0 \\ 21 & -6 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 11 & 3 & 0 \\ 6 & -1 & 1 \end{bmatrix}$$

$$= -\frac{1}{11} \begin{bmatrix} 3 & -1 & 0 \\ -11 & 6 & 0 \\ -25 & 6 & -1 \end{bmatrix} \begin{bmatrix} 0 & 6 & 0 \\ 14 & 18 & 0 \\ 12 & -6 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$14. 11x_1^2 + 6x_1x_2 + 11x_2^2 = 200$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 11 & 3 \\ 3 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 200$$

$$A = \begin{bmatrix} 11 & 3 \\ 3 & 11 \end{bmatrix}$$

$$|\lambda I - A| = 0$$

$$(\lambda-11)^2 - 9 = 0$$

$$\lambda = 4, 18$$

$$\lambda = 4$$

$$\begin{bmatrix} -7 & -3 \\ -3 & -7 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\lambda = 18$$

$$\begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$P = \left[\frac{x_1}{|x_1|}, \frac{x_2}{|x_2|} \right] = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} Y$$

$$\therefore D = \begin{bmatrix} 4 & 0 \\ 0 & 10 \end{bmatrix}$$

$$\therefore Q = [y_1 \ y_2] \begin{bmatrix} 4 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 200$$

$$4y_1^2 + 10y_2^2 = 200$$

24.

$$Q = X^T A X = Y^T D Y = \sum_{n=1}^n \lambda_n y_n^2$$

따라서 $Q > 0$ 이기 위해 λ 는 양수여야 하고

$Q < 0$ 이기 위해 λ 는 음수여야 하고

$Q \geq 0$ ~ $Q \leq 0$ 이려면 양수, 음수를 가질 수 있다.

8.5

$$A = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix} = A^H$$

$$A^H = A \text{ (Hermitian Matrix)}$$

$$|\lambda - A| = 0$$

$$\begin{vmatrix} \lambda - 2 & -i \\ i & \lambda - 2 \end{vmatrix} = 0$$

$$(\lambda - 2)^2 - 1 = 0$$

$$\lambda = 1, 3$$

$$\lambda = 1$$

$$\lambda = 3$$

$$\begin{bmatrix} -1 & -i \\ i & -1 \end{bmatrix} x = 0$$

$$\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} i \\ -1 \end{bmatrix}$$

$$x = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$2. A = \begin{bmatrix} i & H i \\ -1 + i & 0 \end{bmatrix}$$

$$A^H = \begin{bmatrix} -i & -1 - i \\ 1 - i & 0 \end{bmatrix}$$

$$A^H = -A \text{ (Skew-Hermitian)}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} i - \lambda & H i \\ -1 + i & -\lambda \end{vmatrix} = 0$$

$$-\lambda(i - \lambda) - (i+1)(i-1) = 0$$

$$\lambda^2 - i\lambda + 2 = 0$$

$$\lambda = -i, 2i$$

$$\lambda = -i \quad \begin{bmatrix} 2i & H i \\ -1 + i & i \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} -i \\ -1 + i \end{bmatrix}$$

$$\begin{bmatrix} -i & H i \\ -1 + i & -2i \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} H i \\ i \end{bmatrix}$$

$$4. A = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

$$A^H = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}$$

$$A^T = \frac{1}{\det(A)} \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}$$

$$-A = A^H = A^{-1}$$

$\therefore$ Skew-Hermitian, unitary

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda & -i \\ -i & \lambda \end{vmatrix} = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

$$\lambda = -i$$

$$\lambda = i$$

$$\begin{bmatrix} -i & -i \\ -i & -i \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} i & -i \\ -i & i \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$5. A = \begin{bmatrix} -i & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{bmatrix}$$

$$A^H = \begin{bmatrix} i & 0 & 0 \\ 0 & -i & 0 \\ 0 & 0 & i \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{i} & 0 & 0 \\ 0 & -\frac{1}{i} & 0 \\ 0 & 0 & \frac{1}{-i} \end{bmatrix}$$

$$-A = A^H = A^{-1} \text{ (Skew-Hermitian, unitary)}$$

$$\begin{bmatrix} -i & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{bmatrix}$$

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda + i & 0 & 0 \\ 0 & \lambda - i & 0 \\ 0 & 0 & \lambda + i \end{vmatrix} = 0$$

$$(\lambda + i)^2 (\lambda - i) = 0$$

$$\lambda = -i(\text{2x}), i$$

$$\lambda = -i$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -2i & 0 \\ 0 & 0 & 0 \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$x = i$$

$$\begin{bmatrix} 2i & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2i \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

12.

$$A = \begin{bmatrix} 1 & i & 4 \\ -i & 3 & 0 \\ 4 & 0 & 2 \end{bmatrix}$$

$$A^H = \begin{bmatrix} 1 & -i & 4 \\ i & 3 & 0 \\ 4 & 0 & 2 \end{bmatrix}$$

$$\det(A) = 6 - 48 - 2 = -44$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 6 & - & - \\ & -14 & \\ & & - \end{bmatrix}$$

$$A = A^H \quad (\text{Hermitian})$$

$$\bar{x}^T A x = [1 \ -i \ i] \begin{bmatrix} 1 & i & 4 \\ -i & 3 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ i \\ -i \end{bmatrix}$$

$$= [1 \ -i \ i] \begin{bmatrix} -4i \\ 2i \\ 4-2i \end{bmatrix} = -4i + 2 + 4i + 2 = 4$$

$$18. \quad A = A^T = A^H \quad (\text{Hermitian})$$

$$\therefore AA^H = A^2 = A^H A$$

$$-A = A^T = A^H \quad (\text{skew-Hermitian})$$

$$\therefore AA^H = -A^2 = A^H A$$

$$A^{-1} = A^H \quad (\text{unitary})$$

$$\therefore AA^H = AA^{-1} = I = A^{-1}A = A^H A$$

Review

$$3. A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|\lambda I - A| = 0$$

$$(\lambda - a)(\lambda - d) - bc = 0$$

$\therefore \lambda$ 는 2차식 근

$$4. \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

$\therefore$ Real Matrix can have complex eigenvalue

$$\begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^2 - 1 = 0$$

$$\lambda = \pm 1$$

$\therefore$ Complex Matrix can have real eigenvalue

$$12. A = \begin{bmatrix} -7 & 4 \\ -12 & 7 \end{bmatrix} \quad \lambda I - A = \begin{bmatrix} \lambda + 7 & -4 \\ 12 & \lambda - 7 \end{bmatrix}$$

$$|\lambda I - A| = 0$$

$$(\lambda + 7)(\lambda - 7) + 48 = 0$$

$$\lambda^2 - 1 = 0$$

$$\lambda = \pm 1$$

$$\lambda = -1$$

$$\begin{bmatrix} 6 & -4 \\ 12 & -8 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$\lambda = 1$$

$$\begin{bmatrix} 8 & -4 \\ 12 & -6 \end{bmatrix} X = 0$$

$$X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$14. \begin{bmatrix} 7 & 2 & -1 \\ 2 & 7 & 1 \\ -1 & 1 & 8.5 \end{bmatrix}$$

```
>> A = [7 2 -1; 2 7 1; -1 1 8.5];
>> [S,D] = eig(A)
```

S = eigenvector

```
-0.6667    -0.5130    -0.5407
 0.6667    -0.7348    -0.1248
-0.3333    -0.4437    0.8319
```

D = eigenvalue

```
4.5000         0         0
         0     9.0000         0
         0         0     9.0000
```

17.

$$A = \begin{bmatrix} 5 & -2 \\ -16 & -3 \end{bmatrix}$$

$$|\lambda I - A| = (\lambda - 5)(\lambda + 3) - 20 = 0$$

$$\lambda^2 - 2\lambda - 35 = 0$$

$$\lambda = -5, 7$$

$$P = \begin{bmatrix} 2 & -2 \\ 1 & -3 \end{bmatrix}$$

$$P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} -3 & 2 \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{-4} \begin{bmatrix} -3 & 2 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{4} & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} \end{bmatrix}$$

$$\hat{A} = \begin{bmatrix} \frac{3}{4} & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -16 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 1 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{4} & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 8 & -4 \\ -23 & 29 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{24}{4} + \frac{23}{2} & -3 - \frac{29}{2} \\ 2 + \frac{23}{2} & -1 - \frac{29}{2} \end{bmatrix} = \begin{bmatrix} \frac{35}{2} & -\frac{35}{2} \\ \frac{27}{2} & -\frac{31}{2} \end{bmatrix}$$

$$|\lambda I - \hat{A}| = 0$$

$$(\lambda - \frac{35}{2})(\lambda + \frac{31}{2}) - \frac{27 \times 35}{4} = 0$$

$\lambda = 5, 7$ 은 미행하면 성립한다.

$$21. \begin{bmatrix} -12 & 22 & 6 \\ 8 & 2 & 6 \\ -8 & 20 & 16 \end{bmatrix}$$

$$(\lambda I - A) = 0$$

$$\begin{vmatrix} \lambda+12 & -22 & -6 \\ -8 & \lambda-2 & -6 \\ 8 & -20 & \lambda-16 \end{vmatrix} = 0$$

$$x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc$$

$$(\lambda+12)(\lambda-2)(\lambda-16) + 22 \times 6 \times 8 - 8 \times 20 \times 6 + 6(\lambda-2) \times 8 - 120(\lambda+12) - 22 \times 8(\lambda-16) = 0$$

$$\lambda^3 - 6\lambda^2 - 184\lambda + 1384 + 1056 - 960 + 48\lambda - 96 - 120\lambda - 1440 - 176\lambda + 2816 = 0$$

$$\lambda^3 - 6\lambda^2 - 432\lambda + 1710 = 0$$

$$\lambda = -20, 4, 22$$

$$\lambda = -20$$

$$\begin{bmatrix} -8 & -22 & -6 \\ -8 & -22 & -6 \\ 8 & -20 & -36 \end{bmatrix} x_1 = 0$$

$$x_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\lambda = 4$$

$$\begin{bmatrix} 16 & -22 & -6 \\ -8 & 2 & -6 \\ 8 & -20 & -12 \end{bmatrix} x_2 = 0$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\lambda = 22$$

$$\begin{bmatrix} 34 & -22 & -6 \\ -8 & 20 & -6 \\ 8 & -20 & 8 \end{bmatrix} x_3 = 0$$

$$x_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\det(P) = 4 + 1 + 1 - 1 + 2 + 2 = 9$$

$$\therefore P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & -3 \\ 0 & 3 & 3 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & -3 \\ 0 & 3 & 3 \end{bmatrix}$$

$$\therefore D = P^{-1}AP = \frac{1}{9} \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & -3 \\ 0 & 3 & 3 \end{bmatrix} \begin{bmatrix} -12 & 22 & 6 \\ 8 & 2 & 6 \\ -8 & 20 & 16 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ -1 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & -3 \\ 0 & 3 & 3 \end{bmatrix} \begin{bmatrix} -24 - 22 + 12 & -12 + 22 - 6 & -12 + 22 + 12 \\ 16 - 2 + 6 & 8 + 2 - 6 & 8 + 2 + 12 \\ -16 - 20 + 16 & -8 + 20 - 16 & -8 + 20 + 32 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & -3 \\ 0 & 3 & 3 \end{bmatrix} \begin{bmatrix} -40 & 4 & 22 \\ 20 & 4 & 22 \\ -20 & -4 & 44 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & -3 \\ 0 & 3 & 3 \end{bmatrix} \begin{bmatrix} -40 & 4 & 22 \\ 20 & 4 & 22 \\ -20 & -4 & 44 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} -120 & -60 & 0 & 0 \\ 0 & 12+12+12 & 0 \\ 0 & 0 & 44+44+8 \end{bmatrix}$$

$$= \begin{bmatrix} -20 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 22 \end{bmatrix}$$